# 1.1 What is a network?

## In this lesson

You will learn:

- what a computer network is
- why networks are important at school, at home and in the workplace.

**Spiral back**

Last year, you learned about how computers help us learn, work and enjoy our free time. Now you will learn about how computers and other devices are connected and why this is important.

## What are networks for?

In offices and schools, computers are usually connected to make a computer **network**.

Networks are complicated to set up. Networks are expensive to run because they need a trained person to fix problems. Most organisations think the cost is worth it because a network has many advantages.

### The advantages of networks

- **Communicating:** We can send messages and emails over the network.
- **Sharing:** We use a network to share files and expensive devices like printers.
- **Saving work:** We can save files onto network storage drives. You can use your files on any computer on the network. It is easy to share files with other people.
- **Working together:** Networks help people to work together.

## What makes a network?

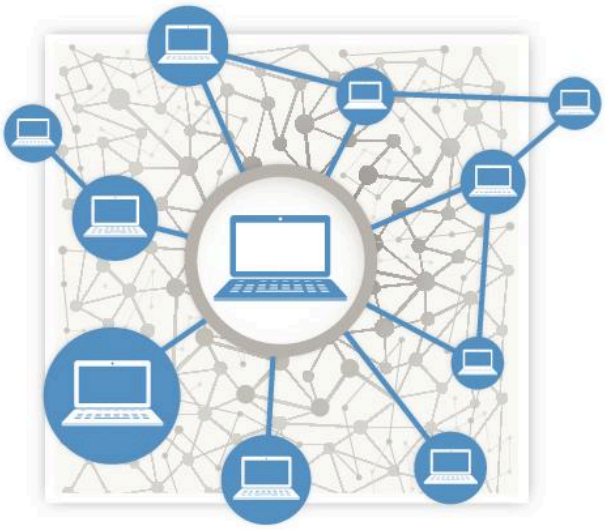
Four things are needed to make a network:

- **Network devices** to send messages and files between computers, and to store files and applications
- **Cables** to join the devices together
- **Network software** to make the devices work together
- Rules that allow all the parts of a network to work together

You will learn more about network devices and how they are connected in Lessons 1.2 and 1.3.



## The two main types of network



- A **local area network** (LAN) joins computers in a single building. A LAN allows people in that building to work together. A school network is a LAN.
- A **wide area network** (WAN) joins computers that are far apart. An organisation with offices in different cities or countries uses a WAN so the employees can work together. The internet is a WAN.

## The 'invisible' network

A school network can have 30 or more network devices. It can have 10 kilometres of cable running around the school.

A network is big, but most people don't notice it is there. Most of the equipment is locked away to keep it safe. But there are clues that the network exists. In this unit you will learn to spot the clues.



### Activity

Web Valley Primary School has 60 network connections. The average cable used for each connection is 50 metres long. How many metres of network cable are there in the school? Convert your answer to kilometres.





### Extra challenge

Use a spreadsheet to solve the problem in the activity.

Then use your spreadsheet to solve this problem:

Web Valley Secondary School has 140 network connections. The average cable is 65 metres long. How many kilometres of network cable are there in Web Valley Secondary School?

### Think again

Do you save your school work to the school network? What is your personal area of the school network called? Is there an area on the network that your teacher uses to share files with the class?

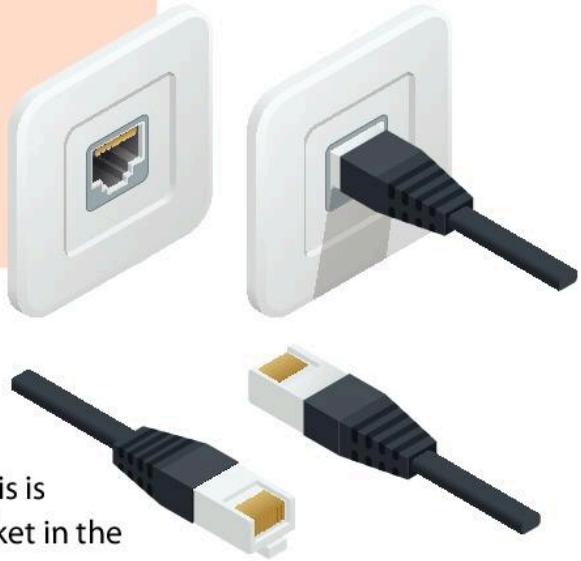
Where is the network equipment in your school stored?

# 1.2 Network connections

## In this lesson

You will learn:

- how computers are connected to a network
- how to set a strong password and how to keep your password safe.



## Connecting to a network

There are two ways to connect a computer to a network.

### Wired connection

If you use a cable to plug the computer into the network, this is called a **wired connection**. A network cable connects a socket in the computer to an identical socket on the wall of a room.

If you see a socket like the one in the picture, it is a clue that the building you are in has a local area network (LAN).

### Wireless (Wi-Fi) connection

You can also connect to a network using a **wireless connection**. This is also called **Wi-Fi**.

Networks use a device called a **wireless access point** (WAP) to provide a wireless connection. If you are close to a WAP, you can connect to a network without using a cable. A place where a wireless connection is available is called a hotspot.

You usually see WAPs high on the wall or on the ceiling.

Laptop and tablet computers usually use a wireless connection to a network. Desktop computers are usually connected to a network with a cable.



## Using a network login

To use a network, you must have a **login**. A login is made up of a user name and a **password**.

Using a login keeps the network safe. Only people who have permission can use the network. Only you can see your own files on the network.




### Keep your password safe

You must keep your password safe to protect your information and your work.

- Never tell anyone your password.
- Change your password regularly.
- Choose a strong password.

### A strong password

A strong password is difficult for another person to guess. You can use the passphrase method to make a strong password.

1. Think of a short phrase that you can remember. Use two or three short words with eight to twelve characters in total. For example: **'Ginger cats'**
2. Remove the spaces between the words: **Gingercats**
3. Replace some of the letters with other characters that look similar. For example: a → @, e → 3, s → $.
4. This is your new strong password: **Ging3rc@t$**



### Activity

Use the passphrase method to make a new strong password for your login.



### Extra challenge

Create a poster that reminds students to keep their password safe. Include a guide for creating a strong password. Use your own example of a strong password. Can you suggest different characters to use instead of letters in the phrase?



### Explore more

Logins are not just used for school networks. People use logins at work and for online shopping and banking. Ask an adult at home to help you make a list of all the things they use logins for.

Share the passphrase method of creating a strong password with the adult.

# 1.3 Network devices

## In this lesson

You will learn:

- more about network devices, software and rules.

## Devices in a network

A network needs special devices. Each device does an important job in the network.

### Servers

A **server** is a powerful computer. A network uses several servers. Each server does a different job.

**Print server:** When you print a picture or a document, the print server makes sure your file is printed correctly.

**File server:** This server makes sure your documents are saved correctly.

**Email server:** This server makes sure your emails reach the right people.



### Switches and hubs

**Switches** are like major junctions in a road traffic system. When you send a message, a switch decides which cable the data needs to travel down to reach the right destination. For example, if you send an email, the switch makes sure your email goes to the email server. A **hub** is a type of switch.

### Routers

A **router** connects a network to the internet. Your school network has a router. If you have an internet connection at home, it uses a small router.



## How do network devices work together?

The devices in a network work together to get a job done. Here is what happens if you send an email from your school to your home.

You press 'Send'.

Your computer sends the email to a **switch**.

The switch sends the email to the **email server**.

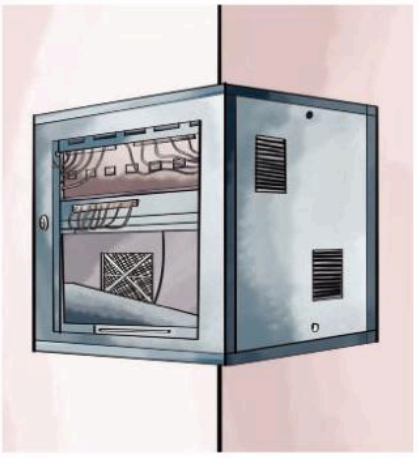
The email server sends the email to the **router**.

The router sends the email to the **internet**.

Your email reaches your home.

## Where are network devices stored?

Network devices are usually stored in a locked room called a **server room**. They are stored in metal cabinets. Sometimes they are stored in small cabinets in classrooms.



## Software and protocols

Two more things are needed to make a network work.

- Software is used to make sure the network runs properly. For example, network software is used to set up user logins.
- All parts of the network must understand each other. The whole network must follow rules so that jobs are done in the correct order. In a network the rules are called **protocols**.



### Activity

Can you find any clues in your school that tell you the school has a network? If you can, take photos or make drawings as evidence.



### Extra challenge

Read the description of what happens in a network when you send an email. Write a similar description of what happens when you print a document.



### Explore more

If you have an internet connection at home, ask an adult to help you find the router. Be careful not to disturb any cables. Can you see anything that you recognise from studying this unit?

# 1.4 The internet

## In this lesson

You will learn:

- what the internet is
- how the internet is different from the world wide web.

## What is the internet?

The **internet** is a wide area network (WAN) that connects computers across the world. It is the biggest network in the world.

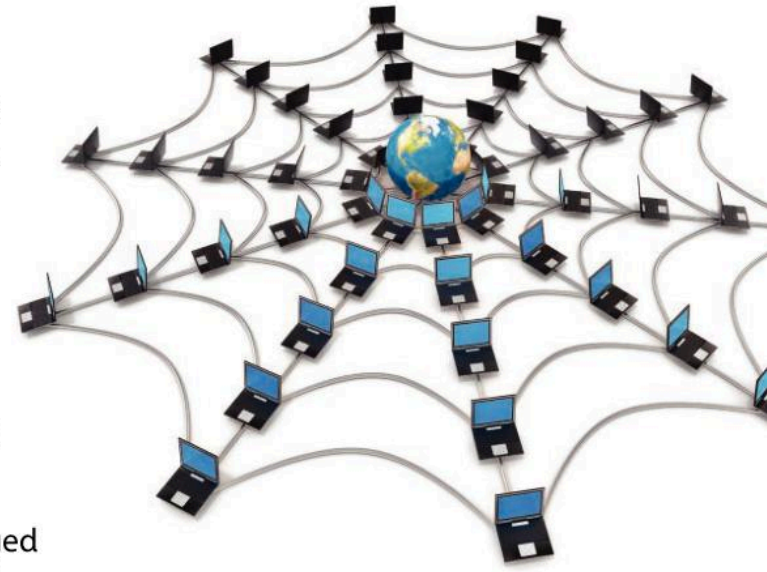
People use the internet to communicate and to share things with other people. The internet helps people to learn and work together.

The internet is not owned or controlled by a single person or organisation.

## Internet services

The internet is a network of connections. Using those connections, we can make use of a range of services.

- **The world wide web:** Made of all the web pages in the world.
- **Email and chat:** Ways of sending messages and content to people you know.
- **Video conferencing:** People across the world can meet as if they were in the same room.
- **Streaming:** Games, music and videos can be played on your local device without being downloaded first.



## The world wide web

The **world wide web** (www or just 'web') is a part of the internet. The web is all the websites and web pages stored on the internet. You use a web browser to look at web pages on the web. You use a search engine to search for information on the web.

The web is a powerful tool because every web page links to other pages. The links on web pages are called hyperlinks. If you drew lines between the hyperlinks on all the web pages on the web, it would look like a huge spider web.

## Computers and the internet

You can use a computer to access all internet services. You use a browser to view the web. You can use other software applications (often called apps) for other internet services such as email.

Many people use a smartphone to access the internet. A smartphone can connect to the internet from anywhere that has a phone signal.

## Smart devices

Many home devices are connected to the internet. These are called **smart devices**.

Home security systems can send a warning to your smartphone if sensors detect sound or movement in your home while you are out. You can check security cameras on your smartphone to see if there is an intruder.

You can use your smartphone to control home air conditioning and heating systems so the temperature is comfortable when you get home. You can switch on lights to make it look as if someone is home.






### Activity

List the main internet services. Say which ones you have used. Give an example.



### Extra challenge

Which devices do you use to connect to the internet? Which services do you prefer to access using a smartphone? Which services do you prefer to access using a computer? Give examples.

### Think again

Research smart home heating or air conditioning systems on the web. List some of the features. How do smart home air conditioning and heating systems help the environment?